Exercise 27.1

A photon counter is pointed at a remote star for one minute, in order to infer the rate of photons arriving at the counter per minute, λ. Assuming the number of photons collected r has a Poisson distribution with mean λ,

$$P(r \mid \lambda) = \exp(-\lambda)\frac{\lambda^r}{r!},$$

and assuming the improper prior $P(\lambda) = 1/\lambda$, make Laplace approximations to the posterior distribution

(a) over λ

(b) over $\log \lambda$. [Note the improper prior transforms to $P(\log \lambda) =$ constant.]

Solution:

Exercise 27.2

Use Laplace's method to approximate the integral

$$Z(u_1, u_2) = \int_{-\infty}^{\infty} \mathrm{d}a \, f(a)^{u_1} \left(1 - f(a)\right)^{u_2},$$

where $f(a) = 1/(1 + e^{-a})$ and u_1, u_2 are positive. Check the accuracy of the approximation against the exact answer (23.29, p. 316) for $(u_1, u_2) = (\frac{1}{2}, \frac{1}{2})$ and $(u_1, u_2) = (1, 1)$. Measure the error $(\log Z_P - \log Z_Q)$ in bits.

Solution:

Exercise 27.3

Linear regression. N datapoints $\{(x^{(n)}, t^{(n)})\}$ are generated by the experimenter choosing each $x^{(n)}$, then the world delivering a noisy version of the linear function

$$y(x) = w_0 + w_1 x,$$

$$t^{(n)} \sim \text{Normal}\left(y(x^{(n)}), \sigma_\nu^2\right).$$

Assuming Gaussian priors on w_0 and w_1, make the Laplace approximation to the posterior distribution of w_0 and w_1 (which is exact, in fact) and obtain the predictive distribution for the next datapoint $t^{(N+1)}$, given $x^{(N+1)}$. (See MacKay (1992a) for further reading.)

Solution: